

22.3 Transforming regressors

Transforming a regressor changes only the mean function. The errors are untouched, and there is no Jacobian, because the data being explained stay the same. What is at stake is bias. If $E(Y) = \mathbf{X}_2\boldsymbol{\beta}_2 + \beta_1 g(\mathbf{x}_1)$ and the analyst fits $\mathbf{x}_1$ itself, the missing part $\beta_1\{g(\mathbf{x}_1) - \mathbf{x}_1\}$ acts as an omitted regressor (Proposition 5.5.7(a) and Theorem 19.1.2). It shows up as curvature in the residuals, and tests against a larger model detect it (Proposition 14.7.1 and Theorem 14.7.2). This section finds the right g , first by looking, then by estimating a power.

22.3.1 Component-plus-residual plots

The component-plus-residual plot of Section 20.1 plots the partial residuals $\hat{\mathbf{e}} + \hat{\beta}_1\mathbf{x}_1$ against $\mathbf{x}_1$. Its least squares line has slope $\hat{\beta}_1$ and residuals $\hat{\mathbf{e}}$, as in the added-variable plot, and that section warns that it can mislead when $\mathbf{x}_1$ is related nonlinearly to the other regressors. The next result measures how far.

Proposition 22.3.1 — What the plot shows

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ have full column rank, with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$, and let $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{e}}$ be the coefficients and residuals of the least squares fit of $\mathbf{y}$ on $\mathbf{X}$. Let $\mathbf{M}$ and $\mathbf{M}_2$ project onto $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}_2)$, and $\tilde{\mathbf{x}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{x}_1$.

- (a) If $E(\mathbf{Y}) = \mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2$, where $\mathbf{g}$ has entries $g(x_{i1})$ for some function g , then

$$E(\hat{\mathbf{e}} + \hat{\beta}_1\mathbf{x}_1) = \mathbf{g} - \mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1), \quad b = E(\hat{\beta}_1) = \frac{\tilde{\mathbf{x}}_1^\top \mathbf{g}}{\|\tilde{\mathbf{x}}_1\|^2}. \quad (22.3.1)$$

- (b) In particular, if $\mathcal{C}(\mathbf{X}_2)$ is spanned by $\mathbf{1}$ alone, the expected partial residuals are $\mathbf{g}$ shifted by a constant, so the plot shows g exactly apart from its level.

Proof.

- (a) Since $\mathbf{X}_2\boldsymbol{\beta}_2 \in \mathcal{C}(\mathbf{X})$, $E(\hat{\mathbf{e}}) = (\mathbf{I} - \mathbf{M})(\mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2) = (\mathbf{I} - \mathbf{M})\mathbf{g}$. By Theorem 6.6.2(a), $\hat{\beta}_1 = \tilde{\mathbf{x}}_1^\top \mathbf{Y} / \|\tilde{\mathbf{x}}_1\|^2$, whose mean is $\tilde{\mathbf{x}}_1^\top \mathbf{g} / \|\tilde{\mathbf{x}}_1\|^2 = b$ because $\tilde{\mathbf{x}}_1 \perp \mathcal{C}(\mathbf{X}_2)$. By Lemma 6.6.1(c), $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{x}}_1\tilde{\mathbf{x}}_1^\top / \|\tilde{\mathbf{x}}_1\|^2$, so $(\mathbf{I} - \mathbf{M})\mathbf{g} = \mathbf{g} - \mathbf{M}_2\mathbf{g} - b\tilde{\mathbf{x}}_1$. Adding $b\mathbf{x}_1 = b\tilde{\mathbf{x}}_1 + b\mathbf{M}_2\mathbf{x}_1$ gives $\mathbf{g} - \mathbf{M}_2\mathbf{g} + b\mathbf{M}_2\mathbf{x}_1$, which is Eq. 22.3.1.
- (b) If $\mathcal{C}(\mathbf{X}_2) = \text{span}\{\mathbf{1}\}$, then $\mathbf{M}_2\mathbf{v} = \bar{v}\mathbf{1}$ for every $\mathbf{v}$. ■

The distortion $\mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1)$ is the part of the *nonlinearity* $\mathbf{g} - b\mathbf{x}_1$ that the other regressors reproduce linearly. Correlation with $\mathbf{x}_1$ is not in itself harmful; a regressor that resembles the curvature is. Cook (1993) showed for random regressors that the plot recovers g , up to a linear term, when the conditional mean of the other regressors given x_1 is linear in x_1 , and proposed CERES plots for other cases. The **augmented partial residual plot** of Mallows (1986) adds $\mathbf{x}_1^2$ to the fit and plots $\hat{\mathbf{e}} + \hat{\beta}_1\mathbf{x}_1 + \hat{\beta}_{11}\mathbf{x}_1^2$. Its expectation is $\mathbf{g}$ minus the part of $\mathbf{g} - b_1\mathbf{x}_1 - b_{11}\mathbf{x}_1^2$ that the other regressors reproduce, small when g is nearly quadratic (Exercise (B3)).

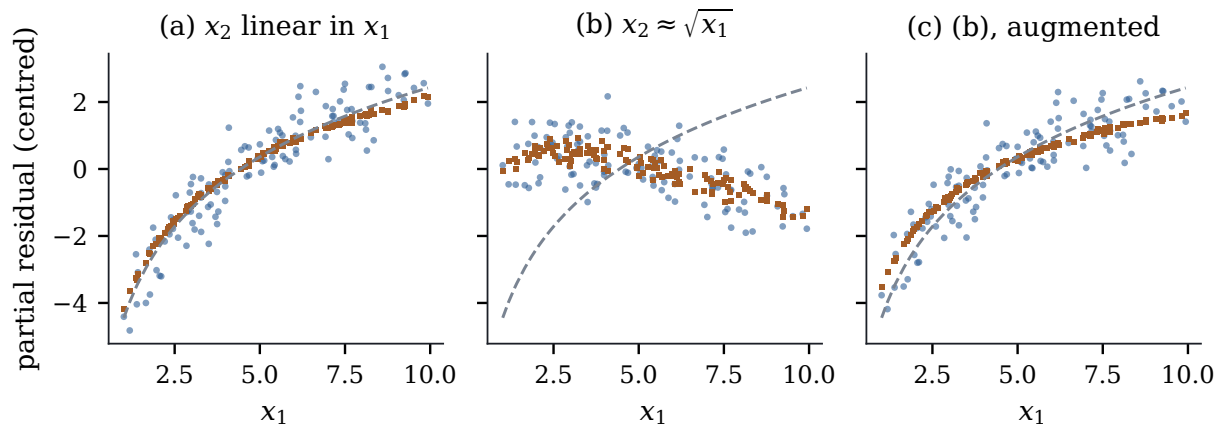


Figure 22.3.1: Component-plus-residual plots for simulated data with $E(Y) = 3 \log x_1 + x_2$. Points: partial residuals. Dashed: the true g , centred. Small squares: the expected partial residuals Eq. 22.3.1, which scatter because they depend on the simulated x_2 . (a) x_2 linearly related to x_1 . (b) x_2 close to $\sqrt{x_1}$: the plot hides the curvature. (c) The augmented plot for the data of (b).

Example. (When the plot misleads)

The script simulates 120 cases with $E(Y) = 3 \log x_1 + x_2$, x_1 uniform on $[1, 10]$, and measures how much of the curvature of g the expected plot keeps (the coefficient of the nonlinear part of g in that of Eq. 22.3.1; 1 is faithful). In panel (a) of Figure 22.3.1, x_2 is linear in x_1 plus noise, with correlation 0.95, and the plot keeps 0.99. In panel (b), x_2 is a noisy square root of x_1 , with correlation 0.99; it resembles the logarithm and takes over much of the curvature, and the plot keeps only 0.45, its slope even changing sign. The augmented plot, panel (c), restores 0.88.

```
import numpy as np

rng = np.random.default_rng(2204)
n = 120
x1 = rng.uniform(1, 10, n)
g = 3 * np.log(x1)                                     # the true shape in x1
standardize = lambda v: (v - v.mean()) / v.std()
x2_lin = standardize(x1) + rng.normal(0, 0.3, n)         # linear in x1, plus noise
x2_sqrt = standardize(np.sqrt(x1)) + rng.normal(0, 0.1, n) # nonlinear in x1, plus
noise

def lstsq(X, v):
    coef, *_ = np.linalg.lstsq(X, v, rcond=None)
    return coef

def partial_residuals(y, x2, augmented=False):
    """e + b1 x1 (+ b11 x1^2 if augmented), from the fit of y on 1, x1 (, x1^2), x2.
    """
    X = np.column_stack([np.ones(len(y)), x1] + ([x1**2] if augmented else []) + [x2])
    coef = lstsq(X, y)
    e = y - X @ coef
```

```

return e + coef[1] * x1 + (coef[2] * x1**2 if augmented else 0)

def curvature_retained(v):
    """Coefficient of the nonlinear part of g in the nonlinear part of v (1 = shape
    kept)."""
    Z = np.column_stack([np.ones(n), x1])
    nonlin = lambda u: u - Z @ lstsq(Z, u)
    return nonlin(v) @ nonlin(g) / (nonlin(g) @ nonlin(g))

noise = rng.normal(0, 0.6, n)
results = {}
for name, x2, aug in [("linear", x2_lin, False), ("square root", x2_sqrt, False),
                     ("square root, augmented", x2_sqrt, True)]:
    mean = g + x2 # E(Y); partial residuals
                  are linear in y
    results[name] = (partial_residuals(mean + noise, x2, aug), partial_residuals(
        mean, x2, aug))
    print(f"{name:23s} corr(x1, x2) {np.corrcoef(x1, x2)[0, 1]:.2f}    curvature
    retained "
          f"{curvature_retained(results[name][1]):.2f}")

```

[Run this code online](#)

22.3.2 Power transformations: Box and Tidwell

When the plot suggests a power, it can be estimated. Box and Tidwell (1962) considered the mean function

$$E(\mathbf{Y}) = \mathbf{X}_2 \boldsymbol{\beta}_2 + \beta_1 \mathbf{x}^{(\alpha)}, \quad \mathbf{x}^{(\alpha)} = (x_1^{(\alpha)}, \dots, x_n^{(\alpha)})^\top, \quad (22.3.2)$$

for a positive regressor $\mathbf{x}$, with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$ and Eq. 22.2.1 applied to the regressor. For fixed α this is a linear model, and $\hat{\alpha}$ minimizes the residual sum of squares $\text{SSE}(\alpha)$ of the fit of $\mathbf{y}$ on $[\mathbf{X}_2, \mathbf{x}^{(\alpha)}]$. Under normal errors it is the maximum likelihood estimate, and $\{\alpha : n \log\{\text{SSE}(\alpha)/\text{SSE}(\hat{\alpha})\} \leq \chi_{1,0.05}^2\}$ is an asymptotic 95% interval under the conditions of Wilks's theorem, which include $\beta_1 \neq 0$ (Exercise (C1)). The derivative of $x^{(\alpha)}$ with respect to α at $\alpha_0 \neq 0$ is $x^{\alpha_0} \log x / \alpha_0 - x^{(\alpha_0)} / \alpha_0$, and at $\alpha_0 = 0$ it is $\frac{1}{2}(\log x)^2$ (Exercise (B1)). Since $x^{(\alpha_0)}$ and $\mathbf{1}$ are already in the model, the part that matters is the **constructed variable**

$$\mathbf{w} = \begin{cases} (x_i^{\alpha_0} \log x_i / \alpha_0)_{i=1}^n, & \alpha_0 \neq 0, \\ (\frac{1}{2}(\log x_i)^2)_{i=1}^n, & \alpha_0 = 0. \end{cases} \quad (22.3.3)$$

For $\alpha_0 = 1$ it is $x \log x$.

Theorem 22.3.2 — Box–Tidwell

Let $\mathbf{X}_0 = [\mathbf{X}_2, \mathbf{x}^{(\alpha_0)}]$ have full column rank p , with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$, let $\mathbf{w}$ be the constructed variable Eq. 22.3.3, and suppose $[\mathbf{X}_0, \mathbf{w}]$ has full column rank $p + 1 < n$. Let $\hat{\beta}_1$ be the coefficient of $\mathbf{x}^{(\alpha_0)}$ and $\hat{\boldsymbol{\varepsilon}}_0$ the residual vector in the least squares fit of $\mathbf{y}$ on $\mathbf{X}_0$, and let

$\hat{\gamma}$ be the coefficient of $\mathbf{w}$ in the fit of $\mathbf{y}$ on $[\mathbf{X}_0, \mathbf{w}]$.

- (a) **Exact test.** If $\mathbf{Y} \sim N_n(\mathbf{X}_0\boldsymbol{\beta}, \sigma^2\mathbf{I})$, which is Eq. 22.3.2 with $\alpha = \alpha_0$, then the t statistic of $\hat{\gamma}$ has the $t(n - p - 1)$ distribution.
- (b) **The step.** If $\hat{\beta}_1 \neq 0$, one Gauss–Newton step for minimizing $\|\mathbf{y} - \mathbf{X}_2\boldsymbol{\beta}_2 - \beta_1\mathbf{x}^{(\alpha)}\|^2$, started from the least squares coefficients at α_0 , moves α to

$$\alpha_1 = \alpha_0 + \hat{\gamma}/\hat{\beta}_1. \quad (22.3.4)$$

- (c) **Stationarity.** $\text{SSE}(\alpha)$ is differentiable at α_0 with $\text{SSE}'(\alpha_0) = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w} = -2\hat{\beta}_1\hat{\gamma}\|(\mathbf{I} - \mathbf{M}_0)\mathbf{w}\|^2$, where $\mathbf{M}_0$ projects onto $\mathcal{C}(\mathbf{X}_0)$. So if $\hat{\beta}_1 \neq 0$, then $\hat{\gamma} = 0$ iff α_0 is a stationary point of SSE, and every fixed point of the iteration Eq. 22.3.4 is a stationary point.

Proof. Write $\mathbf{d} = \partial\mathbf{x}^{(\alpha)}/\partial\alpha$ at α_0 . By the derivative computed above, $\mathbf{d} - \mathbf{w}$ is a multiple of $\mathbf{x}^{(\alpha_0)}$, so $\mathbf{d} - \mathbf{w} \in \mathcal{C}(\mathbf{X}_0)$ and $(\mathbf{I} - \mathbf{M}_0)\mathbf{d} = (\mathbf{I} - \mathbf{M}_0)\mathbf{w} =: \tilde{\mathbf{w}}$, which is nonzero because $[\mathbf{X}_0, \mathbf{w}]$ has full rank. By Theorem 6.6.2(a), $\hat{\gamma} = \tilde{\mathbf{w}}^\top \mathbf{y} / \|\tilde{\mathbf{w}}\|^2$, and $\tilde{\mathbf{w}}^\top \mathbf{y} = \mathbf{w}^\top (\mathbf{I} - \mathbf{M}_0)\mathbf{y} = \mathbf{w}^\top \hat{\boldsymbol{\epsilon}}_0$.

- (a) The column $\mathbf{w}$ depends on $\mathbf{x}$ alone, not on $\mathbf{Y}$. The t test for $\hat{\gamma}$ is therefore the F test of $\mathcal{C}(\mathbf{X}_0)$ against the fixed larger space $\mathcal{C}([\mathbf{X}_0, \mathbf{w}])$, with $t^2 = F$ and a symmetric null distribution. By Proposition 14.7.1(a), $F \sim F(1, n - p - 1)$ when $\mathbf{E}(\mathbf{Y}) \in \mathcal{C}(\mathbf{X}_0)$, so $t \sim t(n - p - 1)$.
- (b) The mean function $\boldsymbol{\mu}(\boldsymbol{\beta}_2, \beta_1, \alpha) = \mathbf{X}_2\boldsymbol{\beta}_2 + \beta_1\mathbf{x}^{(\alpha)}$ has derivative matrix $\mathbf{J} = [\mathbf{X}_2, \mathbf{x}^{(\alpha_0)}, \hat{\beta}_1\mathbf{d}]$ at the starting point, where the residual is $\hat{\boldsymbol{\epsilon}}_0$. The Gauss–Newton step is the least squares coefficient vector of $\hat{\boldsymbol{\epsilon}}_0$ on $\mathbf{J}$, and its last entry is the change in α . By Theorem 6.6.2(a) that entry is

$$\frac{\{(\mathbf{I} - \mathbf{M}_0)\hat{\beta}_1\mathbf{d}\}^\top \hat{\boldsymbol{\epsilon}}_0}{\|(\mathbf{I} - \mathbf{M}_0)\hat{\beta}_1\mathbf{d}\|^2} = \frac{1}{\hat{\beta}_1} \frac{\tilde{\mathbf{w}}^\top \hat{\boldsymbol{\epsilon}}_0}{\|\tilde{\mathbf{w}}\|^2} = \frac{\hat{\gamma}}{\hat{\beta}_1},$$

using $\tilde{\mathbf{w}}^\top \hat{\boldsymbol{\epsilon}}_0 = \tilde{\mathbf{w}}^\top \mathbf{y}$, which holds because $\tilde{\mathbf{w}} \perp \mathcal{C}(\mathbf{X}_0)$.

- (c) Let $\mathbf{X}(\alpha) = [\mathbf{X}_2, \mathbf{x}^{(\alpha)}]$, which has full column rank near α_0 , and let $\hat{\boldsymbol{\beta}}(\alpha)$ be its least squares coefficients, a differentiable function of α . Then $\text{SSE}(\alpha) = \|\mathbf{y} - \mathbf{X}(\alpha)\hat{\boldsymbol{\beta}}(\alpha)\|^2$, and

$$\text{SSE}'(\alpha_0) = -2\hat{\boldsymbol{\epsilon}}_0^\top \{\hat{\beta}_1\mathbf{d} + \mathbf{X}_0\hat{\boldsymbol{\beta}}'(\alpha_0)\} = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{d} = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w},$$

because $\hat{\boldsymbol{\epsilon}}_0 \perp \mathcal{C}(\mathbf{X}_0)$ and $\mathbf{d} - \mathbf{w} \in \mathcal{C}(\mathbf{X}_0)$. Finally $\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w} = \hat{\gamma}\|\tilde{\mathbf{w}}\|^2$ from the first paragraph. ■

Part (a) is exact because this constructed variable is built from the fixed regressor, not from the response as in Proposition 22.2.4. Part (b) identifies the Box–Tidwell iteration with Gauss–Newton, fast near a well-determined minimum and liable to oscillate otherwise, so in one dimension it is safer to minimize $\text{SSE}(\alpha)$ directly; by part (c) the two agree at convergence. Several powers can be stepped at once (Exercise (C2)).

Example. (Transforming income)

With $\log(\text{food})$ as the response and $\text{income}^{(\alpha)}$ as the regressor, starting from $\alpha_0 = 1$, the Box–Tidwell iteration gives -0.053 after one step and -0.182 after two, where it stays. The direct minimization agrees: $\hat{\alpha} = -0.182$, with 95% likelihood ratio interval $[-0.334, -0.036]$. The constructed-variable t statistic is -16.58 at $\alpha_0 = 1$, so untransformed income is decisively rejected, and -2.47 at $\alpha_0 = 0$, beyond the critical value 1.97 , so the logarithm is rejected at the 5% level as well. As a check on part (a), 4000 data sets simulated from the log–log fit give a rejection rate of 0.047 for this test.

The rejection rests largely on one household. The richest, with income 4958 and food expenditure 1827, sits far to the right in panel (c) of Figure 22.1.1, below the line. Without it, $\hat{\alpha} = -0.119$, and the likelihood ratio statistic for $\alpha = 0$ falls from 5.98 to 1.77 . A negative power says food expenditure flattens among the richest households, where the data are few; Cook’s distance (Definition 20.4.1) pursues this.

```
import numpy as np
import statsmodels.api as sm

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
y = np.log(food)

def bc(x, a):
    """Box-Cox transform (x^a - 1)/a, with log x at a = 0."""
    return np.log(x) if abs(a) < 1e-12 else np.expm1(a * np.log(x)) / a

def fit(y, x, a):
    return sm.OLS(y, np.column_stack([np.ones(len(y)), bc(x, a)]).fit()

def box_tidwell_step(y, x, a0):
    """One Box-Tidwell (Gauss-Newton) step: a1 = a0 + gamma/beta1."""
    beta1 = fit(y, x, a0).params[1] # slope of x^(a0) in the null
    fit
    w = x**a0 * np.log(x) / a0 if a0 != 0 else np.log(x) ** 2 / 2 # d bc(x, a)/da,
    up to C(X0)
    aug = sm.OLS(y, np.column_stack([np.ones(len(y)), bc(x, a0), w])).fit()
    return a0 + aug.params[2] / beta1, aug.tvalues[2]

a, path = 1.0, [1.0]
for _ in range(8):
    a, t = box_tidwell_step(y, income, a)
    path.append(a)
print("Box-Tidwell iterates from alpha = 1:", " ".join(f"{v:.4f}" for v in path))
```

[Run this code online](#)

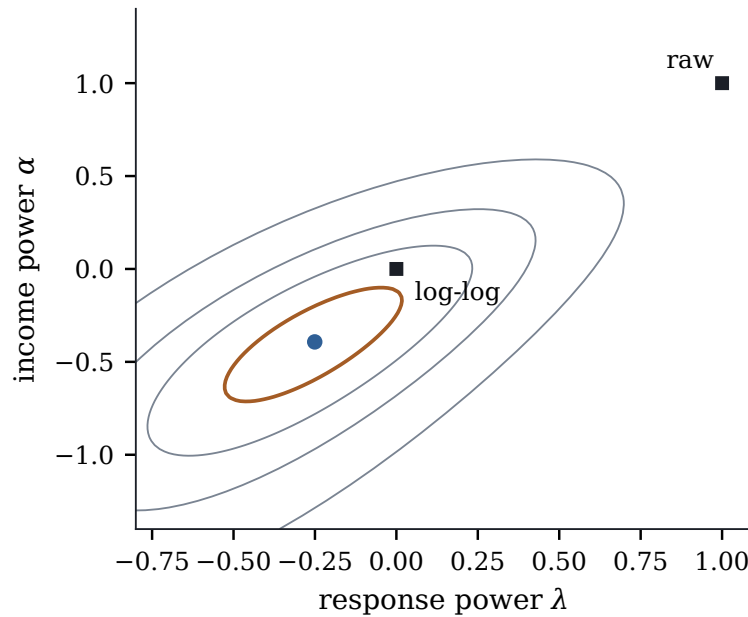


Figure 22.3.2: Joint profile log-likelihood of the response power λ and the income power α for Engel's data. The thick contour bounds the 95% likelihood ratio region, the thin contours are 10, 20 and 40 units of log-likelihood below the maximum (the dot). Squares mark the log-log and raw models.

22.3.3 Choosing both scales together

The best power for the response depends on the scale of the regressor, and conversely, so the two can be estimated together. With response $\mathbf{z}^{(\lambda)}$ and model matrix $[\mathbf{1}, \mathbf{x}^{(\alpha)}]$ the proof of Theorem 22.2.2 goes through unchanged, since the Jacobian involves only $\mathbf{y}$: the joint profile log-likelihood is $-\frac{n}{2} \log \text{SSE}_z(\lambda, \alpha)$ plus a constant.

Example. (Two powers for Engel's data)

Figure 22.3.2 shows the joint profile. The maximum is at $(\hat{\lambda}, \hat{\alpha}) = (-0.25, -0.39)$. The raw scale $(1, 1)$ has likelihood ratio statistic 186.5, and the best pair with untransformed income, $(0.63, 1)$ from Example (Box-Cox for Engel's data), has 167.8: whatever is done to the response, income needs transforming. The log-log pair $(0, 0)$ has statistic 11.2, outside the 95% region, whose cut-off is 5.99 for two parameters. Without the richest household the statistic falls to 6.55, still outside.

The region is a long tilted ellipse, since a lower response power can be partly compensated by a lower income power (Exercise B2). The log-log model lies outside the region (11.2 against 5.99), largely because of the richest household. It is still the natural working choice for interpretation, since its slope is an elasticity, with the caveat that the data reject a constant elasticity and that it may fall among the richest households (Proposition 22.5.3).

22.3.4 Polynomials and other alternatives

Powers suit positive regressors with monotone, smoothly bending effects. Otherwise the alternative is to add columns rather than change one: a polynomial or a spline. These buy flexibility with interpretability, since no single coefficient measures the effect of x (Definition 5.7.1, third qualification), and high-degree polynomials behave badly at the edges of the data. Royston and Altman's (1994) *fractional polynomials* use one or two terms from a small set of powers including the logarithm. Chapter 42 treats polynomial and spline regression.

22.3.5 Exercises

22.3.5.1 A. Check your understanding

Exercise A1.

At $\alpha_0 = 1$ a fit gives $\hat{\beta}_1 = 2.5$, and adding $x \log x$ gives it the coefficient $\hat{\gamma} = -1.5$. What is the next Box–Tidwell value of α ? What does the sign of $\hat{\gamma}$ say about the curvature of the relation?

22.3.5.2 B. Practice

Exercise B1.

Show that $\partial x^{(\alpha)} / \partial \alpha = x^\alpha \log x / \alpha - x^{(\alpha)} / \alpha$ for $\alpha \neq 0$, and that its limit as $\alpha \rightarrow 0$ is $\frac{1}{2}(\log x)^2$. Conclude that in a model containing **1** and $\mathbf{x}^{(\alpha_0)}$ the constructed variable may be taken to be [Eq. 22.3.3](#).

Solution. Differentiate $(x^\alpha - 1)/\alpha$: the quotient rule gives $\{\alpha x^\alpha \log x - (x^\alpha - 1)\}/\alpha^2 = x^\alpha \log x / \alpha - x^{(\alpha)} / \alpha$. As $\alpha \rightarrow 0$, expand $x^\alpha = 1 + \alpha \log x + \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3)$: the numerator is $\alpha \log x + \alpha^2(\log x)^2 - \alpha \log x - \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3) = \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3)$, and dividing by α^2 gives the limit. The term $-\mathbf{x}^{(\alpha_0)}/\alpha_0$ lies in the model space, and removing it changes neither the residualized column $\tilde{\mathbf{w}}$ nor $\hat{\gamma}$.

Exercise B2.

Suppose $y = c x^\beta$ exactly, with $c, \beta > 0$. Show that for $\lambda \neq 0$, $y^{(\lambda)}$ is an affine function of $x^{(\alpha)}$ iff $\alpha = \beta\lambda$, and that for $\lambda = 0$ it is affine in $x^{(\alpha)}$ iff $\alpha = 0$. Explain why data close to a power law give a joint likelihood for (λ, α) with a ridge, and why the slope of the ridge need not be exactly β when the variance also depends on λ .

Solution. For $\lambda \neq 0$, $y^\lambda = c^\lambda x^{\beta\lambda}$, so $y^{(\lambda)} = (c^\lambda x^{\beta\lambda} - 1)/\lambda$, an affine function of $x^{\beta\lambda}$. An affine function of $x^{\beta\lambda}$ is affine in x^α (equivalently in $x^{(\alpha)}$) for all $x > 0$ iff $\alpha = \beta\lambda$, since distinct powers, and a power and the logarithm, are affinely independent functions on $(0, \infty)$. For $\lambda = 0$, $\log y = \log c + \beta \log x$ is affine in $x^{(\alpha)}$ iff $\alpha = 0$, which is again $\alpha = \beta\lambda$. So every pair on the line $\alpha = \beta\lambda$ makes the mean exactly linear, and nearby data make the residual sum of squares small along it. The likelihood also rewards a

constant variance and normal errors, which depend on λ alone, so the maximum lies somewhere along the tilted valley, and its exact direction mixes the two requirements.

Exercise B3.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_1^2, \mathbf{X}_2]$, with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$ and full column rank, and let $E(\mathbf{Y}) = \mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2$. Show that the augmented partial residuals $\hat{\mathbf{e}} + \hat{\beta}_1\mathbf{x}_1 + \hat{\beta}_{11}\mathbf{x}_1^2$ have expectation $\mathbf{g} - \mathbf{M}_2(\mathbf{g} - b_1\mathbf{x}_1 - b_{11}\mathbf{x}_1^2)$, where (b_1, b_{11}) are the expectations of $(\hat{\beta}_1, \hat{\beta}_{11})$. Conclude that the plot is exact, up to a vertical shift, when g is a quadratic.

Solution. Let $\mathbf{Q} = [\mathbf{x}_1, \mathbf{x}_1^2]$ and $\tilde{\mathbf{Q}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{Q}$. As in the proof of [Proposition 22.3.1](#), $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{Q}}(\tilde{\mathbf{Q}}^\top \tilde{\mathbf{Q}})^{-1} \tilde{\mathbf{Q}}^\top$ and the coefficient vector $\mathbf{b}$ of $\mathbf{Q}$ has expectation $(\tilde{\mathbf{Q}}^\top \tilde{\mathbf{Q}})^{-1} \tilde{\mathbf{Q}}^\top \mathbf{g}$. Then $E(\hat{\mathbf{e}}) = (\mathbf{I} - \mathbf{M})\mathbf{g} = \mathbf{g} - \mathbf{M}_2\mathbf{g} - \tilde{\mathbf{Q}}\mathbf{b}$, and adding $\mathbf{Q}\mathbf{b} = \tilde{\mathbf{Q}}\mathbf{b} + \mathbf{M}_2\mathbf{Q}\mathbf{b}$ gives $\mathbf{g} - \mathbf{M}_2(\mathbf{g} - \mathbf{Q}\mathbf{b})$. If $\mathbf{g} = c_0\mathbf{1} + c_1\mathbf{x}_1 + c_2\mathbf{x}_1^2$, then $\mathbf{g}$ lies in $\mathcal{C}(\mathbf{X})$, $\mathbf{b} = (c_1, c_2)^\top$, and $\mathbf{g} - \mathbf{Q}\mathbf{b} = c_0\mathbf{1}$, whose projection is $c_0\mathbf{1}$.

Exercise B4.

In the setting of [Proposition 22.3.1](#), let $\mathbf{X}_2 = [\mathbf{1}, \mathbf{q}]$ with $q_i = x_{i1}^2$, so the second regressor is an exact quadratic in the first, and write $\boldsymbol{\beta}_2 = (\beta_{20}, \delta)^\top$.

- If $g(x) = cx^2$, show that $b = 0$ and that the expected partial residuals are $\mathbf{0}$: the plot is flat, although $E(Y)$ curves in x_1 when $\delta \neq -c$.
- If $g(x) = cx$, show that the plot is exact.
- Explain why no plot against x_1 can tell which of the two regressors carries the curvature in (a).

Solution.

- Here $\mathbf{g} = c\mathbf{q} \in \mathcal{C}(\mathbf{X}_2)$, so $\tilde{\mathbf{x}}_1^\top \mathbf{g} = 0$ and $b = 0$; then $\mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1) = \mathbf{M}_2\mathbf{g} = \mathbf{g}$ and [Eq. 22.3.1](#) is $\mathbf{0}$. The whole of g is the nonlinearity $\mathbf{g} - b\mathbf{x}_1$, and $\mathbf{q}$ reproduces all of it. The mean $E(Y) = \beta_{20} + (c + \delta)x_1^2$ curves unless $c + \delta = 0$.
- Now $b = c\tilde{\mathbf{x}}_1^\top \mathbf{x}_1 / \|\tilde{\mathbf{x}}_1\|^2 = c$, because $\tilde{\mathbf{x}}_1^\top \mathbf{x}_1 = \|\tilde{\mathbf{x}}_1\|^2$, so $\mathbf{g} - b\mathbf{x}_1 = \mathbf{0}$ and the expected partial residuals are $\mathbf{g}$.
- The models $E(Y) = \beta_{20} + cx_1^2 + \delta x_1^2$ with the same $c + \delta$ have the same mean for every case: the split between g and $\mathbf{X}_2\boldsymbol{\beta}_2$ is not identified from the data, so no display of them can recover it.

22.3.5.3 C. Going deeper

Exercise C1.

Suppose $\beta_1 = 0$ in [Eq. 22.3.2](#). Show that the model does not depend on α , so α is not identified. Show that the test of [Theorem 22.3.2\(a\)](#) still has exact size, and argue that its power against changes in α tends to zero as $\beta_1 \rightarrow 0$. What happens to the Box–Tidwell step, and to the likelihood ratio interval for α ?

Exercise C2.

Let $E(\mathbf{Y}) = \mathbf{X}_3\boldsymbol{\beta}_3 + \beta_1\mathbf{x}_1^{(\alpha_1)} + \beta_2\mathbf{x}_2^{(\alpha_2)}$. Show that one Gauss–Newton step from (α_1^0, α_2^0) and the least squares coefficients there changes α_j by $\hat{\gamma}_j/\hat{\beta}_j$, where $\hat{\gamma}_1, \hat{\gamma}_2$ are the coefficients of the two constructed variables in a single regression of $\mathbf{y}$ on the model matrix augmented by both.